

Exercise 2 - E-Commerce Search

Program Output

LINEAR SEARCH
Product Found: Shoes

BINARY SEARCH
Product Found: Shoes

Time Complexity Analysis
Linear Search : $O(n)$
Binary Search : $O(\log n)$

Source Code (Program.cs)

```
using System;

namespace ECommerceSearch
{
    class Product
    {
        public int ProductId;
        public string ProductName;
        public string Category;

        public Product(int productId, string productName, string category)
        {
            ProductId = productId;
            ProductName = productName;
            Category = category;
        }
    }

    class Program
    {
        static Product LinearSearch(Product[] products, int targetId)
        {
            foreach (Product product in products)
            {
                if (product.ProductId == targetId)
                {
                    return product;
                }
            }

            return null;
        }

        static Product BinarySearch(Product[] products, int targetId)
        {
            int left = 0;
            int right = products.Length - 1;

            while (left <= right)
            {
                int mid = (left + right) / 2;

                if (products[mid].ProductId == targetId)
                {
                    return products[mid];
                }

                if (products[mid].ProductId < targetId)
                {
                    left = mid + 1;
                }
                else
                {

```

```

        right = mid - 1;
    }
}

return null;
}

static void Main(string[] args)
{
    Product[] products =
    {
        new Product(101, "Laptop", "Electronics"),
        new Product(102, "Mobile", "Electronics"),
        new Product(103, "Headphones", "Electronics"),
        new Product(104, "Shoes", "Fashion"),
        new Product(105, "Watch", "Accessories")
    };

    int targetId = 104;

    Console.WriteLine("LINEAR SEARCH");

    Product linearResult = LinearSearch(products, targetId);

    if (linearResult != null)
    {
        Console.WriteLine(
            $"Product Found: {linearResult.ProductName}");
    }
    else
    {
        Console.WriteLine("Product Not Found");
    }

    Console.WriteLine();

    Console.WriteLine("BINARY SEARCH");

    Product binaryResult = BinarySearch(products, targetId);

    if (binaryResult != null)
    {
        Console.WriteLine(
            $"Product Found: {binaryResult.ProductName}");
    }
    else
    {
        Console.WriteLine("Product Not Found");
    }

    Console.WriteLine();
    Console.WriteLine("Time Complexity Analysis");
    Console.WriteLine("Linear Search : O(n)");
    Console.WriteLine("Binary Search : O(log n)");
}
}
}

```